

EE675

Assignment – 1

3(a). Kernel Estimation (Implicit through sampling)

The transition kernel represents the probability of moving from one state to another given an action. Instead of explicitly constructing this kernel, which would be computationally expensive, the implementation estimates it implicitly using sampling.

For each state-action pair, the simulator is run multiple times to observe possible next states. These observations approximate the underlying transition probabilities. A larger number of samples provides a better approximation of the true kernel.

3(b). Value Iteration :

Value Iteration is an iterative method used to compute the optimal value function. It works by repeatedly updating the value of each state based on the best possible action.

In this implementation, the expected return for each action is not computed using exact transition probabilities. Instead, it is estimated using multiple samples generated by the simulator. For each state-action pair, K samples are taken, and the average of the observed rewards and next-state values is used to update the Q -value.

The value of a state is then taken as the maximum Q -value over all actions. This process is repeated until the values stabilize. The parameter K controls the accuracy of this estimation.

Q-Function Computation:

The Q-function represents the expected return for taking a specific action in a given state. It is computed using sampled transitions by averaging over multiple simulations.

For each sample, the reward and the value of the next state are combined using the discount factor. The average over all samples gives the estimated Q-value. This approach avoids the need for exact transition probabilities.

Difference Measurement (L1 norm):

To evaluate the quality of the estimation, the difference between two Q-functions is computed using the L1 norm. This is calculated as the sum of absolute differences across all state-action pairs.

This measure provides a simple way to quantify how close the estimated Q-function is to the reference Q-function.

Role of sampling Parameter K:

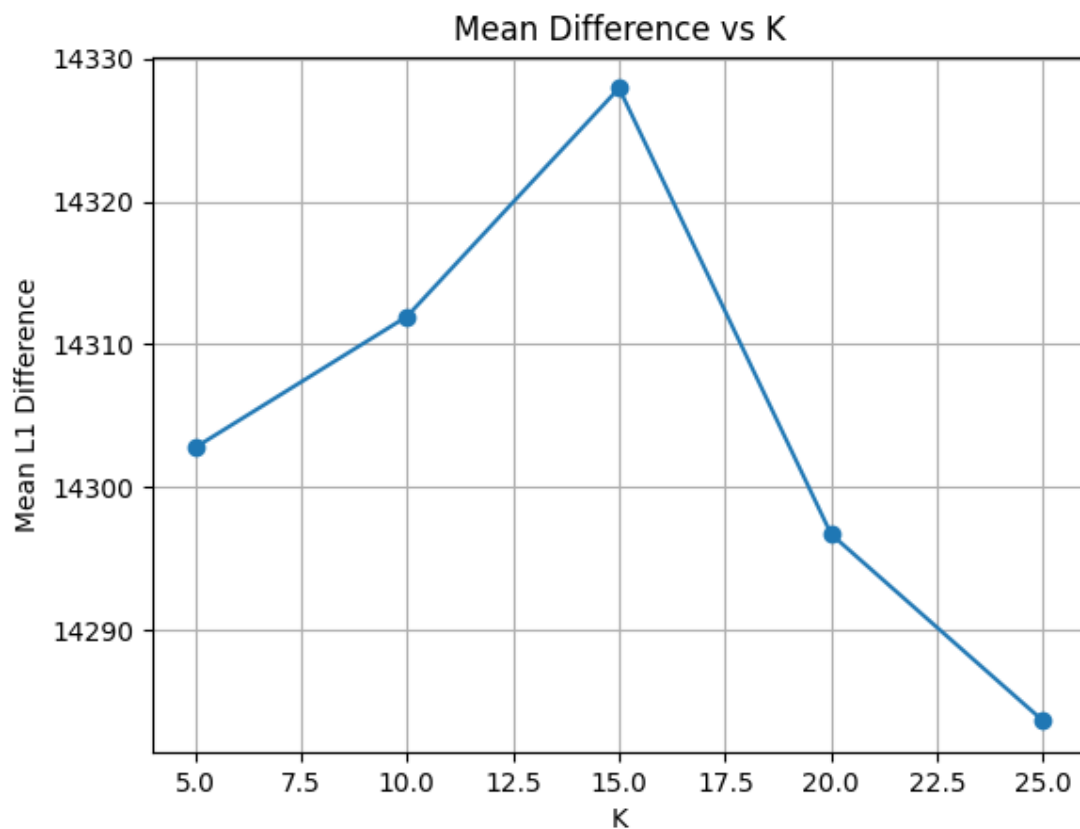
The parameter K controls the number of samples used for estimation. It plays a critical role in balancing accuracy and computational cost.

A small value of K leads to noisy estimates because it relies on fewer samples. A larger value of K improves accuracy by averaging over more samples, but increases computation time.

Overall, these functions work together to demonstrate how sampling can be used to approximate transition dynamics and value functions in large state-space problems.

RESULTS:

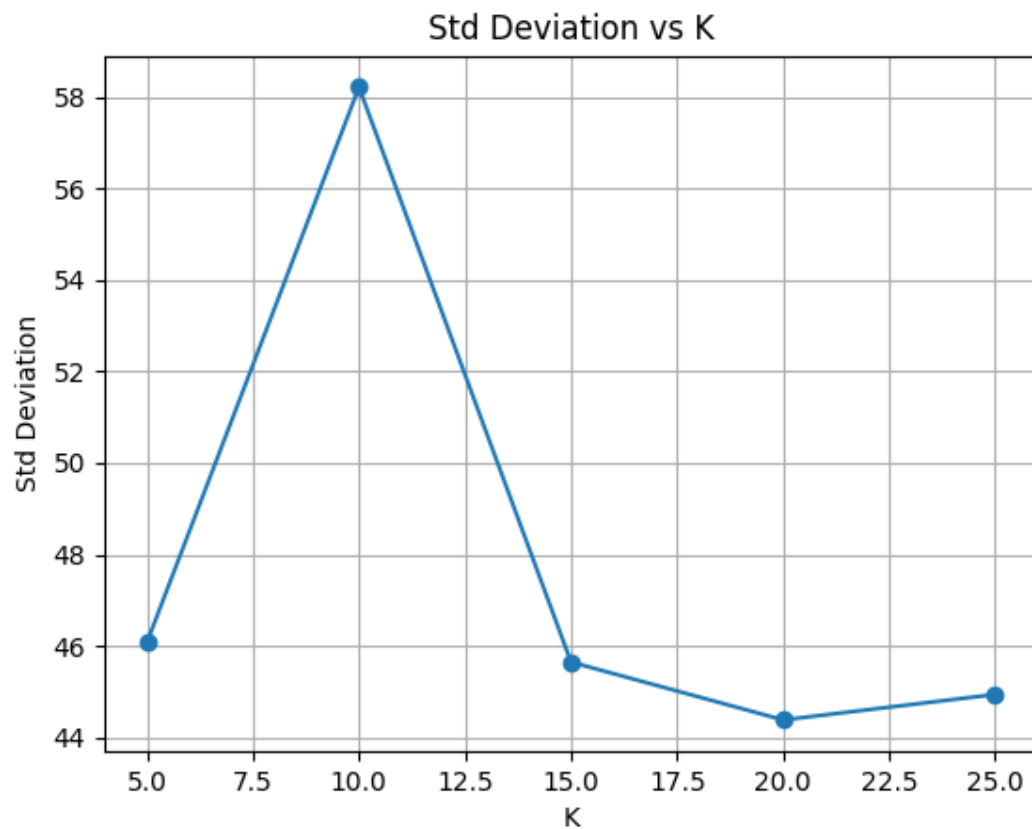
PLOT 1: Mean Difference vs K



The mean L1 difference between the estimated and reference Q-functions shows slight fluctuations for smaller values of K, followed by a decreasing trend for larger K. This behavior is expected because, for small K, the estimation is noisy due to limited sampling. As K increases, the estimates become more accurate, reducing the difference.

PLOT 2: Std Deviation vs K

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The standard deviation decreases as K increases, indicating improved stability of the estimates across different random seeds. With more samples, the effect of randomness is reduced, leading to more consistent results.

CONCLUSION:

Overall, the results demonstrate that increasing the number of samples improves stability and consistency of the estimates, even though the comparison baseline introduces some deviation in the mean difference.